TRAN ANH THUAN

DATA SCIENCE
MACHINE LEARNING ENGINEER

SUMMARY

In general, I have experience working on Data Analyst and Machine Learning with Python in processing data and converting to visualization or text description automatically.

Furthermore, I have mastery of several models in Computer Vision 2D Facial Recognition and 3D Point Cloud, NLP classification and Time-series forecasting.

CONTACT

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Linkedin

Github

Google

ResearchGate

KEY SKILL

Programming: Python, SQL.
Platform: Pytorch, Google Cloud Vertex AI.
3D Point Cloud: Transformer in weakly-supervised point cloud segmentation.
Natural Language Processing: Sentimental classification with SVM, RNN and LSTM, Transformer with BERT.
Time series forecasting: Forecasting stock price by machine learning model Prophet with Backtest Strategy (Backtrader).
Optimization Clustering: K-means, Max Covering Location Problem (MCLP), p-median Problem.

EXPERIENCE

MCSP LAB – PUKYONG NATIONAL UNIVERSITY

2022/03 – PRESENT
Responsibilities: Research Assistant

Point Cloud Segmentation

Given point cloud dataset, we use local graph extraction with Transformer for point cloud with multiple encoder and decoder branches to perform point cloud segmentation tasks.
Responsibilities:

- Introduce Local Geometric Encoding and Local in Global.
- Introduce General Local Graph and Local Spatial Attention.
- Build point cloud segmentation model based on Pytorch and Transformer architecture for point cloud.
- Prove proposed method performance by comparing with state-of-the-art results with large-scale point cloud dataset.

TUAN LOC COMMODITIES

2021/03 – 2022/07
Responsibilities: Data Research Analyst

Accomplishments:

1. Automation Report

Automating monthly report from SQL database to reports as docx and pptx. Programming: Python
Responsibilities:

- Data Cleaning and Processing
- Extract Scatter chart, barchart and twice y-axis chart
- Extract description which compares current data and others such as MoM, YoY

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- Extract report as docx and pptx file with predefined template with corresponding format

2. Algorithmic trading for coffee future market

Building forecasting model and backtest strategy to forecast future price in the coffee market combined with the actual stock.

Programming: Python

Responsibilities:

- Manage and update master data with predefined features
- Data Analyst to filter crucial variable
- Building forecasting model to predict future price
- Building appropriate the backtest strategy
- Compare with current strategies (Momentum, Mean Reverse, Pair Trading,..)

GB AI LAB

2019/09 – 2021/03

Responsibilities: Research Assistant

Accomplishments:

1. Smart Classroom

Building complete solutions for smart classroom applied AI specialized in computer vision including face recognition and emotion detection:

- Gather and structure face data (5 images per student including side face).
- Implement machine learning and deep learning model for face detection.
- Transfer learning by using deep learning models for emotion detection.
- Apply the face recognition technology in automatic attendance and exam verification.
- Apply emotion detection model in evaluating teaching performance.
- Display the valuable information on the dashboard with appropriate charts.

2. A location-based service method for students

We build a clustering algorithm that suggest optimize method from home to school and the school location for planning and visualize results on the map:

- Data Cleaning and Standardization.
- Convert address to corresponding coordinates

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- Build cost function that covers the operating cost and individual cost for changing school location.
- Building algorithm to suggest students change their school and still ensure the school capacity.
- Discover optimized location and suggest it to the government by using location optimization algorithms.

EDUCATION

PUKYONG NATIONAL UNIVERSITY, SOUTH KOREA

2022/03 – Present (Graduation expectation: 2024/02)

MEng, Department of Artificial Intelligence Convergence. Current GPA: 4.3/4.5

Research: 3D Point Cloud segmentation using local graph extraction and Transformer.

UNIVERSITY OF ECONOMICS AND LAW, VNU-HCM

2017/08 – 2021/07

Bachelor of Economics in E-Commerce, Faculty of Information Systems. GPA: 7.8/10. Outstanding student research prize.

Relevant courses: Data Structure and Algorithm, Data analysis with R/Python. Thesis: Design of an AI-based Smart Classroom

TECHNICAL PROJECT

OPINION MINING WITH VIETNAMESE NEWS

2019/10 – 2020/03

Responsibilities: Research Student

Gather news, comments on the Smartphone industry with Vietnamese online newspapers and use deep learning to classify them.

Responsibilities:

- Data Cleaning and Processing
- Feature extraction and vectorizing
- Building several deep learning models and transfer learning (SVM, LSTM, BiLSTM, BERT) for sentiment classification
- Using k-fold to evaluate efficiently

CERTIFICATE

Google Data Analytics Professional Certificate

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IBM Data Science Professional Certificate

Applied Data Science Specialization – University of Michigan

Deep Learning Specialization – Andrew Ng

Third Prize Young Scientist – University of Economics and Law, VNU – 2018 and 2020

Third Prize Goldberg Machine Contest – South Korea – 2017

PUBLICATION

[1] **A.-T. Tran**, H.-S. Le, O.-J. Kwon, S.-H. Lee, and K.-R. Kwon, "General Local Graph Attention in Large-scale Point Cloud Segmentation," in *2023 IEEE International Conference on Consumer Electronics (ICCE)*, 2023, pp. 1–4.

[2] **Anh-Thuan Tran**, Hoanh-Su Le, Suk-Hwan Lee, Ki-ryong Kwon. (2023). Local Graph Point Attention Network in Point Cloud Segmentation. *Under Review.* IEEE Access.

[3] **Anh-Thuan Tran**, Hoanh-Su Le, Suk-Hwan Lee, Ki-ryong Kwon. (2022). Local Graph Transformer in Semantic Point Cloud Segmentation. In The Fall Conference 2022 KMMS (pp. 35-36).

[4] **Tran, A.T.**, Truong, D.H., Le, H.S., Thi, H.X., & Lee, J.h. (2021). A Proposed AI-based Library Usage Management in Smart Universities. 한국경영정보학회 학술대회, 83–91.

[5] **Anh-Thuan, T**. (2021). Design of an AI-based Smart Classroom Management System. (Undergraduate dissertation, University of Economics and Law, Vietnam National University Ho Chi Minh City).

[6] **Anh-Thuan, T.**, Anh-Nhat, N., Xuan-Thanh, B., Nguyen-Tam-An, V., & Hoanh-Su, L. (2020). A proposed method for opinion mining online media of smartphone with Vietnamese text. In The 6th international Conference for Small and Medium Business 2020 (ICSMB 2020) (pp. 362).

[7] **Thuan, T.**, Huy, T., Dung, N., An, V., & Truc, L. (2019). SmartHome IoT Application Development. In The 15th International Conference on Multimedia Information Technology and Application (pp. 371–374).